Konstantin Sobol

Software Engineer | Rust / Backend / Systems / Blockchain

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Professional Summary

Software Engineer with 3 years of practical Rust experience and a strong background in backend, systems-oriented development, algorithms and blockchain applications. Built end-to-end privacy-oriented systems involving smart contracts, tests, blockchain interaction, web and desktop application layers, deployment flow, demos and technical presentations. Strong foundation in Rust semantics, memory management, async/multithreading, C#/C++ development, algorithms, data structures and computer systems.

Core Engineering Profile

- 3 years of practical Rust experience, mainly through smart contracts, blockchain applications and end-to-end DApp projects.
- Built applications across the full development flow: core logic, tests, blockchain interaction, web frontend integration, desktop wallet UI, deployment, demos and presentations.
- Practical experience with privacy-oriented blockchain systems, including ZK-STARK proof generation / verification flow and metadata-hiding communication.
- Strong computer science foundation: C#, C/C++, algorithms, data structures, graph algorithms, computational geometry and computer systems.
- Developed and presented Vara Network / Varathon projects that were recognized with 1st place awards.

Technical Skills

Rust: 3 years of practical experience, primarily through smart contract development, blockchain applications and end-to-end DApp projects. Understanding of move semantics, ownership, borrowing, lifetimes, subtyping, traits, macros, unsafe Rust, async programming, multithreading, atomics, memory management and performance-oriented development. Studied *Programming Rust*, *The Rustonomicon*, *Rust for Rustaceans*, *Rust Atomics and Locks* and *Asynchronous Programming in Rust*.

Blockchain / Web3 / ZK / Privacy: Smart contracts, DApps, wallet flows, blockchain interaction logic, web frontend integration, desktop wallet UI, deployment to blockchain networks, privacy-oriented applications, ZK-STARK proof generation / verification flow, constraints, basic cryptographic primitives and metadata-hiding systems. Studied *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction*.

Backend / Systems: Application architecture, testing, debugging, Git/GitHub, Linux, command-line workflow, CPU architecture, assembly basics, memory hierarchy, virtual memory and processes. Studied *Computer Systems: A Programmer's Perspective*.

C# / C / C++ / Algorithms: 3 years of C# project experience. C/C++ knowledge of OOP, algorithms and data structures, including linked lists, binary trees, binary heaps, B-trees, sorting algorithms and graph algorithms. Experience with Dijkstra algorithm, Yen algorithm, Delaunay triangulation, Voronoi diagrams, computational geometry and complexity analysis.

Python / SQL / Linux: Python basics with Matplotlib, TensorFlow and Keras. SQL basics, relational database design, ER diagrams, schema creation, test data and example queries. Linux basics: files, permissions, shell scripting basics and development environment usage.

Project-Based Experience

Varathon / Gear Foundation / Vara Network

Rust / Blockchain Developer - Project-Based Experience | June 2023 - September 2024

Links: varathon.io | ZK-STARK Mixer: GitHub | Vara Messenger: GitHub / website / slides

- Developed end-to-end blockchain applications on Vara Network, combining smart contracts, tests, blockchain interaction, web and desktop application layers, deployment flow and demos.
- Built privacy-oriented systems including a ZK-STARK cryptocurrency mixer and a metadata-hiding decentralized messenger.
- Worked with Rust code, DApp architecture, wallet flows, privacy-oriented application logic and blockchain integration.
- Prepared technical presentations, demos and project materials for evaluation; projects were recognized with 1st place awards at Vara Network / Varathon hackathons.

Selected Engineering Projects

ZK-STARK Cryptocurrency Mixer Desktop App

Rust / Blockchain / ZK-STARKs / Privacy / Desktop Wallet UI

GitHub: github.com/dan-sobolev-varathon/zk-stark-mixer | Presentation: slides | Demo: video

- Built a privacy-oriented cryptocurrency mixer with desktop wallet UI and native ZK-STARK proof generation.
- Worked with proof generation / verification flow, blockchain interaction logic, smart contracts and tests.
- Implemented a demo-ready wallet flow and prepared technical presentation materials.
- Project was recognized with a 1st place award at a Vara Network / Varathon hackathon.

Mantle Private Messenger

Rust / Blockchain / DApp / Privacy / Web Frontend

GitHub: github.com/kostiksobol/mantle-private-messenger | Website: kostiksobol.github.io/mantle-messenger-site/

- Developed a privacy-preserving blockchain-based messenger project for the Mantle ecosystem.
- Worked on smart contract logic, DApp interaction flow and user-facing web frontend integration.
- Created a dedicated website and maintained a separate codebase for the project.
- Focused on privacy-oriented application logic and metadata-hiding communication concepts.

Additional Engineering Projects

3D Voronoi Diagrams and Delaunay Triangulation

C# / Algorithms / Computational Geometry / Visualization

Video: YouTube demo

- Implemented Delaunay triangulation in 2D and 3D using a divide-and-conquer approach with O(N log N) complexity.
- Built Voronoi diagram construction from Delaunay triangulation in linear time.
- Worked with computational geometry, geometric data structures and algorithmic visualization.
- Visualized results using Matplotlib.

Optical Network Routing Simulation

C# / Graph Algorithms / Simulation / Research-Based Implementation

GitHub: github.com/kasobol/Project_Sobol_Main_Algoritm | Paper: arXiv

- Implemented adapted Dijkstra and Yen algorithms for constrained routing in optical networks.
- Modeled data transmission under realistic network constraints and compared algorithmic efficiency across simulation scenarios.
- Based the implementation on academic paper research.

Blablacar Database Project

SQL / Database Design

GitHub: github.com/kostiksobol/Blablacar_Database

- Designed a relational database model for a Blablacar-like service.
- Created an ER diagram, flowchart, database schema and test data.
- Implemented example SQL queries and basic database usage scenarios.

Academic Project-Based Work

National Research Nuclear University MEPhI

C# / Algorithms Developer - Project Work | September 2019 - June 2021

- Developed long-term algorithmic projects in C#, including computational geometry and optical network routing simulations.
- Implemented Delaunay triangulation, Voronoi diagram construction, constrained graph routing algorithms and research-based simulation logic.
- Worked with algorithmic complexity, data structures, visualization and comparison of algorithmic efficiency.

Education

National Research Nuclear University MEPhI, Moscow

Institute of Intelligent Cybernetic Systems - Applied Mathematics and Computer Science - Incomplete higher education, 2023

Languages & Links

Russian: Native | **English:** B2 / Upper-Intermediate

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